PYTHON'S MAGIC TOOLBOXES

Modules & The Standard Library

Grade 4 | Session 1

PLANRS Academy

The Secret of the Spice Box

Think about a **Masala Dabba** in your kitchen. It is a special toolbox for cooking! [cite: 4, 5]

Visual: A colorful Indian Masala Dabba with various spices.

- It has all the spices you need to cook tasty food! [cite: 6]
- You don't grow spices every time you want to eat; you just take what you need from the box! [cite: 7]

In Python, we have something exactly like this called a **Module**. [cite: 8, 9]

What is a Python Module?

A module is a **special toolbox** in Python that contains ready-made tools (code) to help us build amazing things. [cite: 9, 10]

Just like the Masala Dabba, we only 'import' the tools we need for our specific recipe (or program). [cite: 11]

Meet Our Friends:

- **Aarav** loves numbers and uses the `math` module to solve big problems! [cite: 12]
- **Diya** uses modules to measure her garden! [cite: 45]

How to use 'import'

To use a module, we use the magic word: **import**. [cite: 16, 17]

```
import math
print(math.sqrt(16)) # Finding the square root!
```

This line brings the entire `math` toolbox into your program! [cite: 20, 21]

The Math Module

The `math` module helps us with special calculations that would be hard to do by ourselves. [cite: 25, 26]

For example, Arjun uses `math.pi` to calculate the perfect area of a **Rangoli circle**! [cite: 27, 28]

Visual: A child creating a geometric Rangoli pattern on the floor. [cite: Page 5 Illustration]

The Random Module

The `random` module is all about making choices and having fun! [cite: 29, 30]

Gully Cricket Toss: Ever argued about who bats first? Use `random.choice()` for a digital coin toss! [cite: 31, 32, 33]

Activity 1: Gully Cricket Toss

Let's code a program to pick the first batter! [cite: 34, 35]

```
import random
players = ['Aarav', 'Diya', 'Arjun']
print(random.choice(players), "bats first!")
```

Try it: Add your best friends' names to the list! [cite: 39]

Activity 2: Rangoli Garden

Diya needs to find the side length of a square garden with an area of 81 square meters. [cite: 40, 41]

```
import math
area = 81
side = math.sqrt(area)
print("The side length is", side, "meters")
```

The Standard Library

Python comes with many, many toolboxes already built-in. This collection is called the **Standard Library**. [cite: 46, 47]

The School Bus Analogy:

It is like a **First Aid Kit** in every Indian school bus. You don't need to build it; it is just there ready to help you anytime! [cite: 48, 49]

Quick Quiz!

1. What is a Python module? [cite: 55]
2. How do we bring a module into our program? [cite: 57]
3. Name one thing the `math` or `random` module can do! [cite: 59]

Great job today! See you in Session 2 for more magic! ✨ [cite: 62]